The Modern SQL Challenge

Voltkart · Session 1 — Advanced SQL for Modern Data Engineering

Background

Voltkart is a fast-growing online electronics and appliances retailer in India, selling everything from gaming laptops to air conditioners across the country.

Voltkart has scaled from a niche store to millions of orders, and the data team is feeling the strain. The nightly job still rebuilds reporting tables with full reloads, which has become slow and fragile. At the same time, the analytics team keeps asking for reports the current setup struggles to produce — running revenue trends, per-category rollups across a deep product hierarchy, customer-retention streaks, and team performance up the sales org.

Leadership wants two things from the data team: reporting that answers real questions, and a **reliable incremental load** that only touches what changed instead of rebuilding everything each night.

The challenge

Aarti, Voltkart's Data Engineering lead, is bringing on engineers who can do modern SQL properly — not just SELECTs, but window analytics, recursive hierarchies, and change-driven loading with MERGE. She has prepared a challenge on a copy of the Voltkart schema and wants you to answer a set of analytical questions **and** build the incremental and change-data-capture load logic, all in T-SQL on SQL Server.

What you'll work with

A small relational schema: orders and order-line facts, a daily staging batch of new and changed orders, customers, a product catalogue on a recursive category tree, a sales org chart, and a change-data-capture feed for products. The full data dictionary is in document 04.

> **Note** Everything runs on SQL Server using SSMS. Several questions ask you to read the actual execution plan, so keep "Include Actual Execution Plan" turned on as you work.